

MEC Defense: Securing Smart Stadium Critical Infrastructure

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Context

- Modern high-density venues evolve into massive "Smart Environments" that require real-time monitoring and control
- Tens of thousands of attendees simultaneously connected, generating massive volumes of network traffic
- By using the MEC architecture, data is processed locally which reduces latency and increases privacy
- Public and external traffic should have little to no effect on the sensitive local or private services (e.g. Security video processing)



Key Challenges

- Deployed mobile networks providing internet to both end users and critical infrastructure mean a security failure in one can compromise the other
- Mixed traffic can create security blind spots, requiring careful traffic segregation
- High traffic volume masks malicious activity, making threat detection increasingly more difficult
- External threats may also pose a risk to the internal systems
- Lack of centralized visibility of network activity leaves operators without situational awareness



Goals

- Develop an open source software based security framework deployable on MEC networks
- Implementation of a Security Operations Center (SOC)
- Facilitate the network owner/analyst side of the security process and take real-time decisions based on multiple factors with automatic alert and mitigation, if possible
- Traffic segmentation and isolation
- Redirect and contain threats for forensic analysis



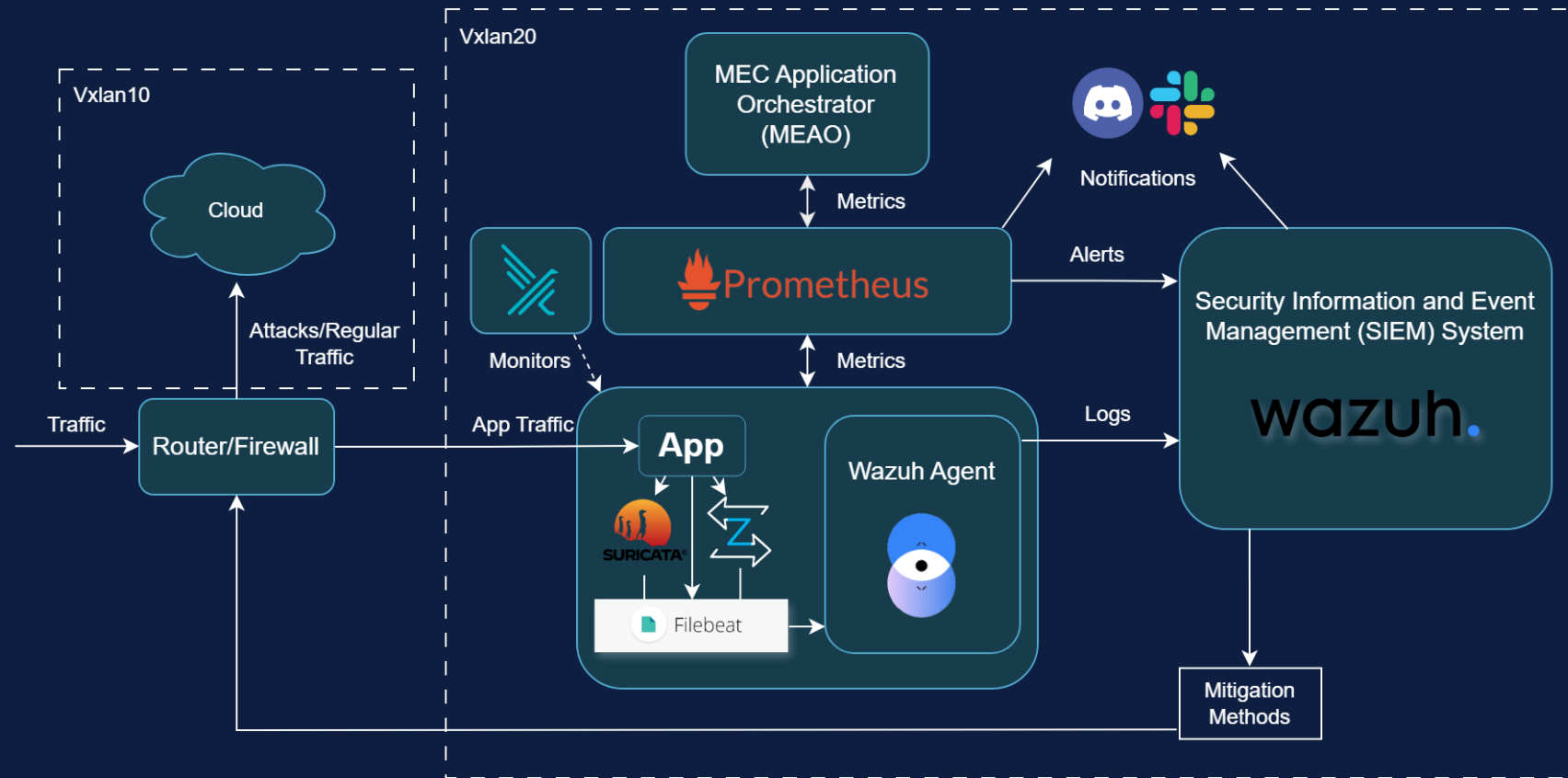


Methodology

Inception	• Setup the MEC environment • Configure a Security Information and Event Management (SIEM) Software • Identify the best way to get the most information out of the network
Elaboration	• Implement log collection and sharing software solutions • Create a demo app to simulate video traffic
Construction	• Collection of MEC application performance metrics • Identificate threats through log patterns • Notify through communication platforms in case of a critical event
Transition	• Implement container security software • Network segmentation • Create a required log standard applied to MEC apps



Architecture



App – Prometheus – SIEM

Suricata– Filebeat – Agent – SIEM

Zeek – Filebeat – Agent – SIEM

App – Filebeat – Agent – SIEM

Falco – Filebeat – Agent – SIEM



Features

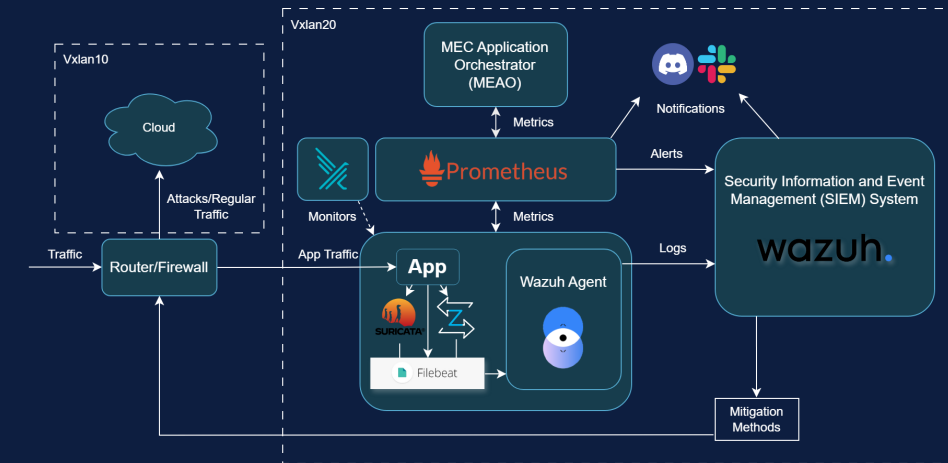
Functional Security Information and Event Management Software
(Wazuh)

Each app has its own agent and collecting tools

Logs collected from the app's interface by Zeek and Suricata and
normalized and grouped in Filebeat

Cluster metric collection and analysis with Prometheus

Non-whitelisted communications are alerted to the network manager





Features

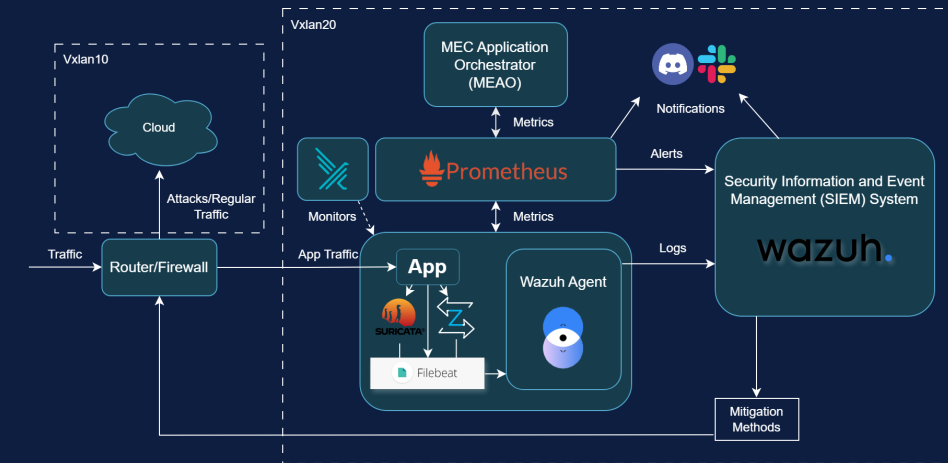
Customized notification rules for Zeek, Falco, Suricata, Prometheus and Wazuh connected to Discord

Video processing app that serves as a reference implementation for standard log procedures

Container runtime security with Falco and whitelisted IPs for inter-app communication

VXLAN based traffic segmentation

Automated mitigation via Router/Firewall triggered by Wazuh alerts





DEMO VIDEO



Future Work

Mitigation methods

Report automation for future forensic analysis

Test the system breaking points

Automate cross referencing between metrics and logs